

Review retrospective:
*the third cumulant — a compound quotient rule, and a
half-wrong dead-open flag*

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Abstract

A soundness review of `Laplace/OneD/AnharmonicThirdCumulant.lean`: the third connected cumulant $\kappa_3 = \partial_h^2 \langle u \rangle_h|_0 = (G_3 G_0^2 - 3G_1 G_2 G_0 + 2G_1^3)/G_0^3 = \langle u^3 \rangle_0 - 3\langle u^2 \rangle_0 \langle u \rangle_0 + 2\langle u \rangle_0^3$ ($u = -(tx)$). Result: **a clean fidelity pass, no edits**, plus one precise report-only note — a dead open `MeasureTheory`, which the reviewer flagged *together with* a load-bearing open `Topology` ($\mathcal{N}$ notation), so “both unused” would have broken the build.

1 Setting

The κ_3 rung of the anharmonic cumulant ladder: the second h -derivative of the perturbed mean $M(h) = G_1(h)/G_0(h)$, sibling of the κ_2 susceptibility.

2 What was reviewed

The single public theorem `anharmonic_third_cumulant`.

3 Fidelity / soundness findings

Sound; GPT found no mismatch. The proof differentiates the susceptibility $S = (G_2 G_0 - G_1^2)/G_0^2$ once more (a compound quotient rule on the partition derivatives). With $N = G_2 G_0 - G_1^2$, $D = G_0^2$, $N' = G_3 G_0 - G_1 G_2$ (using $\partial_h G_n = G_{n+1}$), $D' = 2G_0 G_1$:

$$\frac{N'D - ND'}{D^2} = \frac{(G_3 G_0 - G_1 G_2)G_0^2 - (G_2 G_0 - G_1^2)2G_0 G_1}{G_0^4} = \frac{G_3 G_0^2 - 3G_1 G_2 G_0 + 2G_1^3}{G_0^3},$$

so the -3 and $+2$ coefficients are right. The first derivative of the mean is the susceptibility (`anharmonic_mean_hasDerivAt_general`, eventually on `ball 0 1`); the `iteratedDeriv_succ/iteratedDeriv_one/hev.deriv_eq/hS.deriv` chain takes `iteratedDeriv2 M 0` to κ_3 ; and the docstring’s $\langle u^3 \rangle - 3\langle u^2 \rangle \langle u \rangle + 2\langle u \rangle^3$ is the standard third cumulant, equal to the G_n form by moment-normalisation $\langle u^n \rangle = G_n/G_0$.

4 Simplifications applied

None. GPT’s (B) findings were proof golf (the `hball/hh` ball-is-a-neighbourhood boilerplate, an inlinable `hden_ne`, a G_n abbreviation, a tighter `hS` algebra) — declined — plus one dead-open flag that was *half wrong*: GPT reported “`open MeasureTheory Topology` unused”, but a grep shows `Topology` is load-bearing ($\mathcal{N}$ appears in `hev’s` $=^f [\mathcal{N} 0]$). Only `open MeasureTheory` is genuinely dead. The safe edit is `open MeasureTheory Topology` $\rightarrow$ `open Topology`, not removing both. This single cosmetic one-off dead-open was *not landed* (a worktree + affirmation is not worth one dead word; cf. the `qd` dead-opens deferred to a batch) — it is a precise report-only note for a hygiene pass. File left byte-identical.

5 Disagreements and open questions

None on the mathematics. The dead-open is report-only.

6 Lessons

- **A multi-name open line needs per-name grep verification — a reviewer’s “these opens are unused” can be half right.** GPT lumped a load-bearing `open Topology` (it supplies the $\mathcal{N}$ neighbourhood notation used in the eventual-equality step) with a genuinely-dead `open MeasureTheory` on the same line. Acting on “both unused” would have removed $\mathcal{N}$ and broken the build. This is the `GaussianIntegral` symbol-grep lesson generalised to a shared `open` line: confirm each name’s usage independently (and the build is the final arbiter) before trimming.

7 Follow-ups

- Hygiene (optional): trim `open MeasureTheory Topology` to `open Topology`.
- The cumulants-from-log-partition and general- h susceptibility files complete this arc’s review.